

Customer Churn Prediction

Business Objective:

Predicting customer churn using customer demographic, service usage, contract, payment, tenure, and billing information, so that businesses can identify customers who are at risk of leaving and support customer-retention activities.

Data Set Details:

The dataset contains customer-level telecommunications information, including customer demographics, tenure, services, contract details, payment methods, monthly charges, total charges, and churn-related information.

The project will use the available customer attributes as input features and carefully exclude target/leakage-related fields when building the predictive model.

Acceptance Criterion:

Need to develop and evaluate machine learning models for customer churn prediction and deploy the final end results using Rshiny/Streamlit/Flask or a similar application framework.

Milestones:

30 days to complete the Project

Milestone	Duration	Task	Start - End Date
Kick off and Business Objective discussion	1 day	Project understanding, dataset identification and objective finalization	17-09-2026
EDA	1 Week	Data understanding, data cleaning, exploratory analysis and feature analysis	24-09-2026
Model Building/ Evaluation	1 Week	Feature engineering, model building, evaluation and hyperparameter tuning	01-10-2026
Deployment	1 Week	Deploy the churn prediction application using Streamlit/Flask/Rshiny	08-10-2026
Final presentation	1 week	Final project documentation, presentation and project demonstration	15-10-2026

Protocols:

- 1) All participants should adhere to agreed timelines and timelines will not be extended.
- 2) All the documentation – Final presentation and R/python code to be submitted before the final presentation day.
- 3) All the participants must attend review meetings.